Hackathon Task: URL Shortener with Analytics

Problem Statement

Build a full-stack URL Shortener application where users can create short links for long URLs and track basic analytics such as click count, creation date, and recent visits. The platform should allow authenticated users to manage their links and view performance insights. The goal is to test practical full-stack engineering skills including routing, database modeling, API creation, analytics tracking, and frontend dashboard development.

Mandatory Features

Authentication

- User signup and login
- Protected dashboard routes
- Each user should only manage their own shortened URLs

URL Shortening

- User can submit a long URL and generate a short URL
- Generated short code should be unique
- Clicking the short URL should redirect to the original URL
- Validate that input is a proper URL before shortening

User Dashboard

- View all created short URLs
- Show:
    - Original URL
    - Short URL
    - Created date
    - Total clicks
- Ability to delete a shortened URL
- Ability to copy short URL easily from UI

Analytics

- Count number of clicks per short URL
- Record timestamp of each visit
- Show analytics page/details for each shortened URL
- Display at least:
    - Total click count
    - Last visited time
    - Recent visit history

UI Requirements

- Responsive interface
- Clean dashboard layout
- Proper loading, success, and error states
- Form validation messages

Bonus Features

Custom alias for short URL

- QR code generation for each short URL
- Expiry date for links
- Geolocation or device/browser analytics
- Charts for daily click trends
- Public stats page
- Edit destination URL
- Bulk URL shortening via CSV
- Deployment with live demo

Tech Constraints

- **Frontend:** React
- **Backend:** Node.js (Express / Nest JS)
- **Database:** MongoDB or PostgreSQL
- Use REST APIs
- Use server-side redirect handling
- Use environment variables for configuration
- Passwords must be hashed
- Analytics data should be stored in the database
- Proper backend validation is mandatory
- Do not use an external URL shortening service as the core logic
- UI libraries are allowed if documented

Evaluation Instructions

- Use AI code generation tools to write clean, modular JS code
- Good looking user interfaces will be valued highly
- Follow proper AI app building workflow to build the complete application and have clear documentation of the steps like below:
    - Planning the app
    - Listing and documenting all the features that the app will perform
- Include a `README.md` file with:
    - Setup instructions.
    - Assumptions made.
    - AI Planning document and Architecture diagram of your application.
    - A Loom or YouTube video link explaining and demonstrating your application. **(If you don't submit an explanatory video, your submission will not be reviewed)**
    - Add a line "This project is a part of a hackathon run by https://katomaran.com " in your `README.md` file's bottom
- Submit your github repository in the google form before **12 PM on Thursday, May 21st, 2026**
- Include sample output from the video file: logs, images, and DB entries.
- Usage of AI tools is encouraged but you must completely understand your generated code. The prompts you used will be evaluated during the interview, so please be prepared accordingly.
- You will be asked to run the code and explain the application in your final round of interview
- Although we would recommend everyone to follow the instructions above, you may decide to ignore a few for an innovative reason, but you must ensure your submission is good so you get a chance in the next round to explain it to us.

Best of Luck!